

Cyclic di-AMP drives developmental cycle progression in *Chlamydia trachomatis*

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Junghoon Lee, Scot P Ouellette 

Department of Pathology, Microbiology, and Immunology, College of Medicine, University of Nebraska Medical Center, Omaha, United States

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eLife Assessment

In this **valuable** study, ectopic expression and knockdown strategies were used to assess the effects of increasing and decreasing Cyclic di-AMP on the developmental cycle in *Chlamydia*. The authors **convincingly** demonstrate that overexpression of the *dacA-ybbR* operon results in increased production of c-di-AMP and early expression of the transitionary gene *hctA* and late gene *omcB*. Whilst these results are intriguing, the model currently proposed is over-simplified and likely **incomplete**.

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Abstract

The obligate intracellular bacterium *Chlamydia* alternates between two functional forms during its developmental cycle: elementary body (EB) and reticulate body (RB). However, the molecular mechanisms governing the transitions between these forms are unknown. Here, we present evidence cyclic di-AMP (c-di-AMP) is a key factor in triggering the transition from RB to EB (i.e., secondary differentiation) in the chlamydial developmental cycle. By overexpressing or knocking down expression of c-di-AMP synthase genes, we made strains producing different levels of c-di-AMP, which we linked to changes in secondary differentiation status. Increases in c-di-AMP resulted in an earlier increase in transcription of EB-associated genes, and this was further manifested in earlier production of EBs. In contrast, when c-di-AMP levels were decreased, secondary differentiation was delayed. Based on these data, we conclude there is a threshold level of c-di-AMP needed to trigger secondary differentiation in *Chlamydia*. This is the first study to identify a mechanism by which secondary differentiation is initiated in *Chlamydia* and reveals a critical role for the second messenger signaling molecule c-di-AMP in this process.

Introduction

Chlamydia species are major pathogens of humans and animals. These obligate intracellular bacteria share one key feature: their unique developmental cycle (See ref.(1)). During this cycle, *Chlamydia* transitions between two different functional and morphological forms: the elementary body (EB), an infectious but non-dividing cell, and the reticulate body (RB), a dividing but non-infectious cell (Fig. 1A). A third form, the intermediate body (IB) is a transitional form from the RB to EB. Besides these characteristics, EBs and RBs differ in other ways. For example, EBs are small (~0.3 µm), have a highly disulfide-crosslinked outer membrane(2) and have DNA condensed by histone-like proteins(3). In contrast, RBs are larger (~1 µm), have a Gram-negative cell envelope that lacks peptidoglycan(4–6), and have a dispersed chromosome. RBs divide by an asymmetric polarized division mechanism dependent on MreB-directed peptidoglycan synthesis specifically at the septum(7–10). Not surprisingly, *Chlamydia* expresses genes in a temporally defined manner that correspond broadly with its developmental cycle(11–13). “Early” genes (e.g., *euo*) are expressed immediately upon entry into a target host cell and are likely involved in establishing the intracellular niche of *Chlamydia*, the inclusion, and mediating primary differentiation from EB to RB. “Mid” cycle genes (e.g., *mreB*, *clpPX*) facilitate RB replication and division and inclusion growth. “Late” genes (e.g., *omcB*) are expressed when secondary differentiation is initiated to trigger EB formation. Although developmental gene expression has been characterized for decades, the signals or events that initiate differentiation from one form to the other are not known.

In a 2013 study, another defining characteristic of EBs and RBs was identified: their relative levels of the second messenger signaling molecule cyclic di-AMP (c-di-AMP) (See ref. (14)). Barker et al. studied how IFN β production is activated in cells infected with *C. trachomatis*. The authors identified a role for the innate immune response protein, STING, which recognizes c-di-AMP. *Chlamydia* encodes a diadenylate cyclase enzyme, DacA, associated with c-di-AMP production, that had not been characterized. As part of their study, the authors determined that c-di-AMP accumulates over the course of the chlamydial developmental cycle, that EBs have high levels of this molecule whereas RBs have low levels, and that DacA is a diadenylate cyclase. It is unlikely that *Chlamydia* produces c-di-AMP only to signal host immune responses. Rather, the parsimonious interpretation is that *Chlamydia* uses this signaling molecule to regulate some aspect of its physiology and that activation of host signaling is “accidental” – similar to activation of NOD2 by chlamydial peptidoglycan(15). However, no direct function of c-di-AMP in chlamydial biology has been defined.

Diverse functions of c-di-AMP have been reported in Gram-positive bacteria. For example, c-di-AMP is implicated in the response to changes in osmotic pressure. When the extracellular solute level is high, bacteria prevent dehydration by importing both extracellular solute molecules and cations(16). In these mechanisms, c-di-AMP binds to proteins associated with K⁺ uptake systems such as Ktr/Trk, KimA, Kup, and Kdp and inhibits their activities in *Bacillus subtilis*, *Staphylococcus aureus*, and *Lactococcus lactis*(17–20). Similarly, K⁺ export mechanisms are also regulated by c-di-AMP. In *S. aureus*, the cation/proton antiporter A (CpaA) is activated by binding c-di-AMP(21). Moreover, c-di-AMP binds to the riboswitch upstream of the genes encoding the K⁺ transporters and controls their transcriptional levels(22). With these mechanisms, osmotic stress is controlled by c-di-AMP. Of note, *Chlamydia* lacks annotated orthologs of K⁺ transporters. In addition to osmotic homeostasis, c-di-AMP has been reported to affect DNA replication and sporulation(23). When DNA damage is detected, DisA, a diadenylate cyclase, forms a DNA repair complex with RecA and Rada, resulting in inhibition of diadenylate cyclase activity in *B. subtilis*. Subsequently, c-di-AMP levels decrease, and DNA replication and sporulation are arrested(23–26).

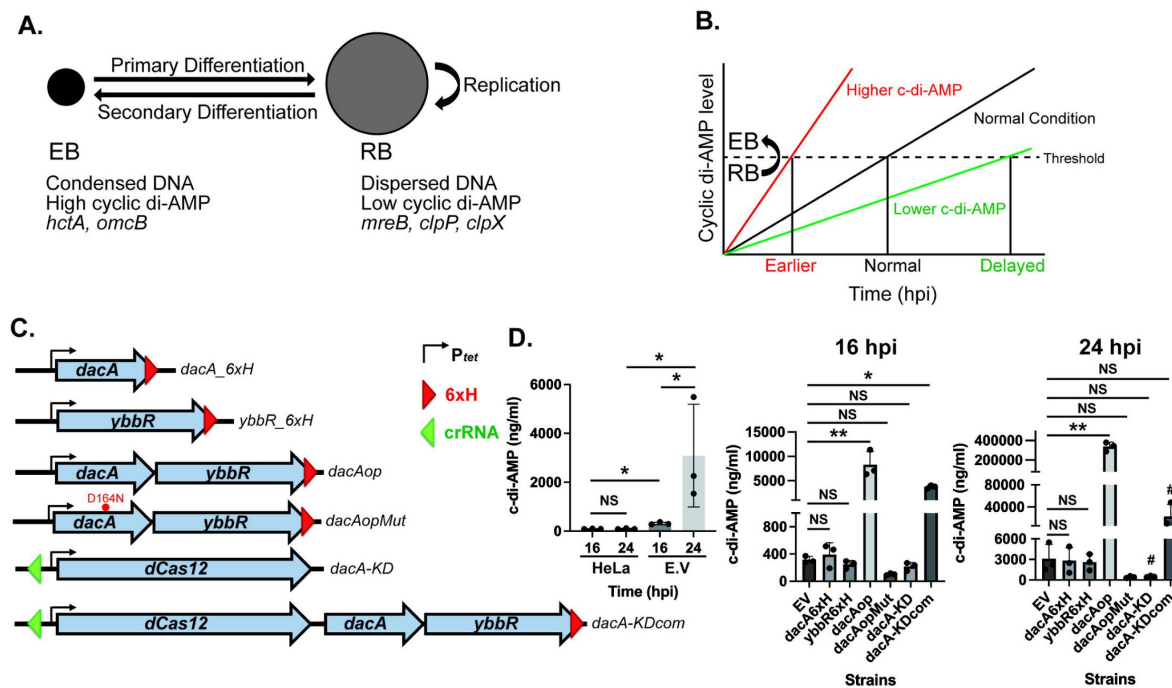


Figure 1.

Cyclic di-AMP accumulation is linked to secondary differentiation in *C. trachomatis*.

A. Defining characteristics of chlamydial EBs and RBs. **B.** A hypothetical model representing the correlation between c-di-AMP levels and secondary differentiation. The dashed line represents a threshold level of c-di-AMP needed to drive secondary differentiation in a given RB. **C.** A schematic diagram of the constructs used in this study. All constructs used are anhydrotetracycline (aTc)-inducible as shown by the P_{tet} promoter. The location of the 6xH tag is shown as well as the approximate location of the crRNA for the CRISPRi vectors. Diagram is not to scale. **D.** Measurement of c-di-AMP concentrations in infected cell lysates from the strains shown in panel C. *C. trachomatis* serovar L2 (434/Bu) transformed with the indicated constructs were infected into HeLa cells. At 10 hpi, expression of the construct was induced or not with 5 nM aTc, and the infected cells were harvested at 16 or 24 hpi. Levels of c-di-AMP in the supernatant were measured using ELISA. The left panel shows the increase from 16 to 24 hpi during overexpression of the control mCherry protein. The middle and right panels show the levels of c-di-AMP in the indicated strains at 16 and 24 hpi, respectively. EV: Empty vector *: p<0.05, **: p<0.001, NS: Not significant via one way ANOVA test. #: p<0.05 via ratio paired t-test as compared to the mCherry control.

Given the differences in c-di-AMP levels between EBs and RBs and its function as a diffusible second messenger signaling molecule, we hypothesized that the accumulation of c-di-AMP during the developmental cycle might be a trigger for secondary differentiation in *Chlamydia* (Fig. 1B). Once a threshold concentration of c-di-AMP has been reached in a given RB, it will begin the differentiation process to an EB. We predicted that, if our hypothesis were correct, then we should be able to alter the levels of c-di-AMP in the organism and affect its differentiation status accordingly. For example, if we increase c-di-AMP production, then we anticipate prematurely triggering EB production. Conversely, if we prevent c-di-AMP production, then we anticipate delaying EB production.

There are three principal mechanisms to regulate c-di-AMP levels: through synthesis by diadenylate cyclase, through degradation by a phosphodiesterase (PDE), and through secretion by a transporter (27). Interestingly, *C. trachomatis* only encodes the synthesis mechanism as it possesses the genes for diadenylate cyclase (*dacA*) and its regulator (*ybbR*) within a bicistronic operon, but no annotated PDEs or c-di-AMP transporters (28). Therefore, to test our hypothesis, we genetically manipulated the levels of DacA and/or YbbR using recently developed strategies in the field. In the present study, we characterized the growth and developmental cycle state of chlamydial strains producing high or low levels of c-di-AMP. Our data show that higher levels of c-di-AMP are directly linked to increased transcript levels of late genes associated with secondary differentiation as well as the concomitant production of EBs at an earlier stage in the developmental cycle. In contrast, in cells with reduced c-di-AMP levels, the developmental cycle was significantly delayed. Based on these data, we conclude that there is a threshold level of c-di-AMP necessary to trigger secondary differentiation in *C. trachomatis*. This is the first study to identify a physiological function for c-di-AMP in *Chlamydia* as well as a mechanism by which secondary differentiation is initiated in these unique bacteria.

Results

Both DacA and YbbR are crucial for cyclic di-AMP synthesis in *C. trachomatis*

To test the link between c-di-AMP levels and chlamydial developmental cycle progression, we made a collection of *C. trachomatis* strains carrying anhydrotetracycline (aTc)-inducible constructs (Fig. 1C). These included strains to overexpress DacA_6xH, YbbR_6xH, both DacA and YbbR_6xH (*dacAop*), or both the catalytically dead DacA(D164N) and YbbR_6xH (*dacAopMut*). As a control for overexpression, we used a strain carrying the same plasmid backbone with aTc-inducible mCherry. For all of these overexpression strains, we emphasize that the chromosomal expression of wild-type *dacA* and *ybbR* is maintained. In addition to the overexpression constructs, we also made a conditional knockdown construct for *dacA* (*dacA-KD*), targeting its promoter region, to decrease the expression of *dacA-ybbR* using a dCas12/crRNA-based system that our group developed for *Chlamydia* (29, 30). Finally, we made a complementation construct for *dacA-KD* by introducing *dacA-ybbR_6xH* (*dacA-KDcom*) 3' to dCas12 such that induction of dCas12 results in the coexpression of DacA and YbbR_6xH as well during knockdown of endogenous *dacA-ybbR*.

Each of the transformant strains were used to infect HeLa cells at a multiplicity of infection (MOI) of 1 and subsequently induced or not with 5 nM aTc at 10 hours post-infection (hpi). We collected cell lysates at 16 and 24 hpi and measured c-di-AMP levels by ELISA (Fig. 1D). The former timepoint is characterized by predominantly an RB population whereas the latter timepoint is characterized by ongoing secondary differentiation and a mixture of RBs, IBs, and EBs. We initially assessed the “empty vector” mCherry expressing control strain for its c-di-AMP levels and observed that c-di-AMP levels were significantly higher at 24hpi compared to 16hpi (Fig. 1D), in agreement with the Barker et al. study indicating higher levels of c-di-AMP in EBs (14). As expected, only basal amounts of c-di-AMP were detected in uninfected HeLa cells (Fig. 1D).

Interestingly, c-di-AMP levels were comparable between the vector control strain and the DacA_6xH or YbbR_6xH overexpression strains, implying that expression of either alone does not alter endogenous c-di-AMP levels. However, when DacA and YbbR_6xH were co-overexpressed, c-di-AMP levels increased by approximately 30-fold at 16 hpi and 120-fold at 24 hpi compared to that of the control strain (Fig. 1D). These data suggest that, in *Chlamydia*, DacA activity requires YbbR for efficient c-di-AMP production. Of note, the levels of c-di-AMP at 16hpi in the *dacAop* strain were higher even than the control strain at 24hpi. To confirm whether DacA enzyme activity is critical for the high levels of c-di-AMP in the *dacAop* strain, we substituted the catalytic residue of DacA with a mutation to render it inactive (D164N) (14). When DacA(D164N) and YbbR_6xH were co-overexpressed (*dacAopMut*), c-di-AMP levels were reduced compared to the control and did not increase between 16 and 24hpi, suggesting that c-di-AMP synthase activity was blocked in this strain. For the *dacA*-KD strain, the c-di-AMP level decreased about 1.2 fold at 16 hpi and ~10 fold at 24 hpi. The loss of c-di-AMP production in this strain was restored in the complemented *dacA*-KD_ *dacAop* strain, which showed a phenotype similar to the *dacAop* overexpression strain. Based on these data, we conclude that both DacA and YbbR are crucial for c-di-AMP synthesis in *Chlamydia*. Importantly, our collection of strains that are high or low producers of c-di-AMP give us an opportunity to test our overarching hypothesis (Fig. 1B).

Overexpression of DacA isoforms, but not YbbR, is detrimental to the chlamydial developmental cycle

DacA and YbbR are both predicted to contain transmembrane domains (3 for DacA and 1 for YbbR) (Fig. S1, (14)). To observe the localization of DacA and YbbR, we infected HeLa cells with transformants encoding *dacA*_6xH or *ybbR*_6xH alone and induced expression of the constructs at 10 hpi with 5 nM aTc. At 24 hpi, infected cells were fixed, and we performed an indirect immunofluorescence assay (IFA) by labeling the major outer membrane protein (MOMP) and 6xH. We observed that both DacA and YbbR localize at the membrane as expected (Fig. 2A and E). Interestingly, when DacA_6xH was overexpressed, the inclusion size appeared qualitatively reduced in size with apparently larger individual organisms than those in the uninduced control. This suggests that overexpressing DacA_6xH alone may negatively impact chlamydial replication.

To investigate this in more detail, we measured EB progeny production using an inclusion forming unit (IFU) assay and genomic DNA levels (a proxy for total bacteria, i.e., EB+RB) by qPCR. To quantify IFUs, a lysate from a primary infection is prepared and used to infect a fresh monolayer of cells. Any inclusions in the secondary infection are derived from an EB present in the primary infection.

The vector control strain showed no differences in IFUs when overexpressing mCherry (Fig. S2). At 24 hpi, both IFUs and genomic DNA levels were significantly decreased in the DacA_6xH overexpression strain compared to that of the uninduced control (Fig. 2B and C), indicating DacA_6xH overexpression disrupts developmental cycle progression. Based on these data, we conclude overexpressed DacA_6xH is detrimental to the chlamydial developmental cycle progression even though it does not impact overall c-di-AMP production (Fig. 1D). To further investigate this, we quantified transcripts by RT-qPCR for a canonical early-cycle gene, *euo*, and two late-cycle genes, *hctA* (an “early” late gene (31)) and *omcB* (a canonical late gene (12)). Consistent with the gDNA and IFU data, overexpression of DacA_6xH resulted in elevated *euo* transcripts and a reduction in the amounts of the late gene transcripts at 24hpi (Fig. 2D). We obtained similar results when overexpressing wild-type DacA without a 6xHis tag, indicating these effects are not linked to the tag (Fig. S3). We also performed similar experiments with the inactive isoform of DacA, DacA(D164N)_6xH. Like overexpression of the wild-type DacA_6xH, overexpression of DacA(D164N)_6xH also negatively affected IFU production while reducing c-di-AMP levels (Fig. S4), similar to the *dacAopMut* strain (Fig. 1D).

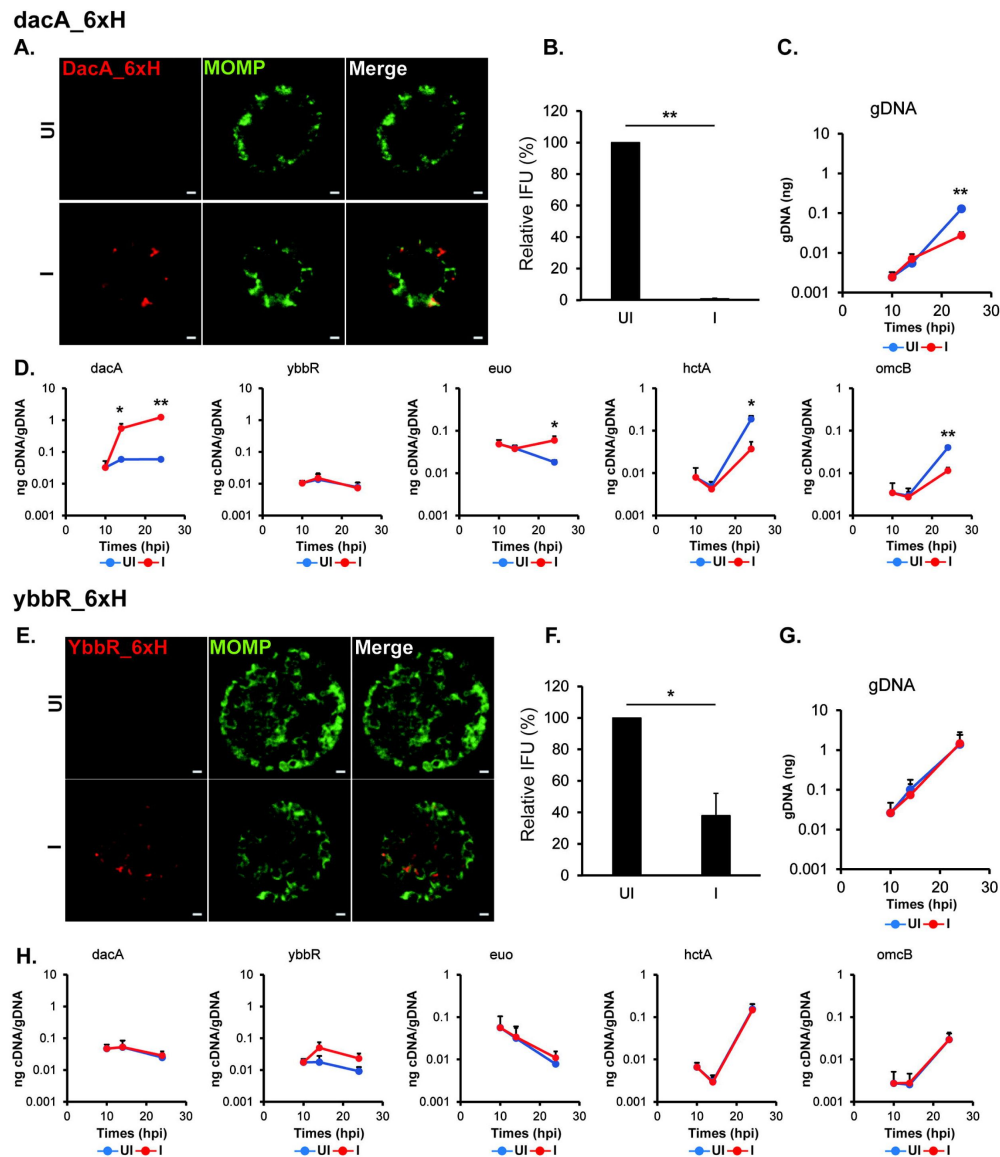


Figure 2.

Overexpression of DacA_6xH, but not YbbR_6xH, is detrimental to the chlamydial developmental cycle.

(A-D) HeLa cells were infected with *C. trachomatis* transformed with a plasmid encoding an anhydrotetracycline (aTc)-inducible DacA_6xH. At 10 hpi, expression of the construct was induced or not with 5 nM aTc, and DNA and RNA samples were collected at 10, 14, and 24 hpi. Immunofluorescence analysis (IFA) and inclusion forming units (IFU) samples were collected at 24 hpi. For IFA images, the green color represents chlamydial major outer membrane protein (MOMP), which is showing the chlamydial cell morphology, and the red color represents DacA or YbbR tagged with six histidine. **A.** IFA images of the *dacA_6xH* strain at 24 hpi. Shown are individual panels of a representative inclusion for DacA_6xH and MOMP labeling as well as the merged image. **B.** Quantification of IFUs from uninduced and induced samples at 24 hpi. **C.** Quantification of genomic DNA by qPCR in uninduced and induced samples. **D.** Quantification of transcripts by RT-qPCR for *dacA*, *ybbR*, *euo*, *hctA*, and *omcB* from uninduced and induced samples. **(E-H)** HeLa cells were infected with *C. trachomatis* transformed with a plasmid encoding an aTc-inducible YbbR_6xH. At 10 hpi, expression of the construct was induced or not with 5 nM aTc, and IFA, IFU, DNA, and RNA samples were collected as above. **E.** IFA images of *ybbR_6xH* strain at 24 hpi. **F.** Quantification of IFU from uninduced and induced samples at 24 hpi. **G.** Quantification of genomic DNA by qPCR in uninduced and induced samples. **H.** Quantification of transcripts by RT-qPCR for *dacA*, *ybbR*, *euo*, *hctA*, and *omcB*. IFA images were acquired on a Zeiss AxioImager.Z2 equipped with an Apotome2 using a 100X lens objective. Scale bar: 1 μ m. UI = uninduced (i.e., -aTc); I = induced (i.e., +aTc) for all sample types. *: $p < 0.05$; **: $p < 0.001$ via two sample equal variance t-test.

We next measured the effect of YbbR_6xH overexpression on the developmental cycle using the same parameters. Unlike DacA_6xH, we measured no difference between uninduced and induced samples for genomic DNA level, indicating no impact of YbbR_6xH overexpression on RB replication or total bacterial numbers (**Fig. 2G**). YbbR_6xH overexpression did result in a statistically significant, ~2-fold decrease in EB production (**Fig. 2F**). This reflects one division cycle difference from the uninduced control and, in the absence of any other phenotypic discrepancy, is not considered biologically relevant. This was further supported by the absence of any effect of YbbR_6xH overexpression on transcript levels for *euo*, *hctA*, or *omcB* (**Fig. 2H**). Therefore, we conclude YbbR_6xH overexpression alone does not affect the chlamydial developmental cycle.

A low level of c-di-AMP decreases the transcript levels of late genes

As mentioned above, the *dacA*-KD strain displayed a lower c-di-AMP level compared to that of the control strain at 24 hpi (**Fig. 1D**). To observe effects of knockdown on the developmental cycle, we examined if there were qualitative defects in inclusion or bacterial morphology by IFA. When the knockdown system was induced, bacterial cell size appeared somewhat enlarged, and we observed the inclusion size appeared smaller compared to that of the uninduced control (**Fig. 3A**). In addition, we also measured the amount of EB progeny (IFUs) and genomic DNA. Although genomic DNA levels were the same, IFUs decreased by approximately 80% compared to that of the uninduced sample at 24 hpi (**Fig. 3B** and **C**). Based on these data, we conclude that low levels of c-di-AMP are detrimental for secondary differentiation. To further explore this, we performed RT-qPCR to quantify transcripts of relevant gene targets. When dCas12 was induced, both *dacA* and *ybbR* transcript levels decreased (**Fig. 3D**). Since *dacA* and *ybbR* are transcribed in an operon, this result is not surprising. Consistent with reduced EB yields, transcripts of the late genes *omcB* and *hctA* were decreased at 24 hpi. In contrast, transcripts of the early gene *euo* were slightly elevated at 24 hpi. We have previously observed no effects on gDNA levels or transcription of these genes when overexpressing the dCas12 gene alone (30), indicating these effects are specific to *dacA*-KD. These data support the conclusion that reducing the levels of c-di-AMP delays developmental cycle progression.

We next tested whether the *dacA*-KD strain's phenotype is complemented by ectopic expression of *dacA-ybbR_6xH* (*dacA*-KDcom) (**Fig. 1C**). This construct encodes not only the *dacA*-knockdown system but also the *dacA-ybbR_6xH* operon as a transcriptional fusion with dCas12. Induction of dCas12 and YbbR_6xH was confirmed by IFA under inducing conditions (**Fig. 3E**). We expected that co-expressing DacA and YbbR_6xH would complement the phenotype of *dacA*-KD. However, the inclusion and bacterial size phenotypes were not fully restored to the uninduced control. We then measured gDNA levels and IFUs and observed that the complemented strain showed an approximate 2-fold reduction in both gDNA and EBs at 24 hpi (**Fig. 3F** & **G**). We next measured transcript levels as described previously. Both *dacA* and *ybbR* transcripts were increased compared to the *dacA*-KD strain (**Fig. 3H**), and transcript levels for these genes were increased beyond the "wild-type" uninduced control levels, suggesting that the complemented strain is more similar to the *dacA*op strain (**Fig. 1D**). Nonetheless, transcripts for the early gene *euo* were identical to those in the uninduced strain (**Fig. 3H**). Surprisingly, *hctA* transcripts were increased ~10-fold at 14 hpi compared to that of the uninduced sample, and *omcB* transcripts were slightly, but not significantly, increased at 14 and 24 hpi (**Fig. 3H**). Based on these unusual findings of elevated late gene transcripts earlier in the developmental cycle, we conclude that overexpressed DacA and YbbR_6xH may alter the timing of secondary differentiation.

High levels of c-di-AMP induce late gene expression

To clarify the effect of DacA and YbbR_6xH overexpression on secondary differentiation, we next evaluated the phenotype of the *dacA*op strain. When we induced expression of this construct, we confirmed the induction of DacA and YbbR_6xH and their colocalization at the membrane (**Fig. 4A**; Pearson correlation coefficient of 0.713 ± 0.109 from 20 inclusions measured by JACoP

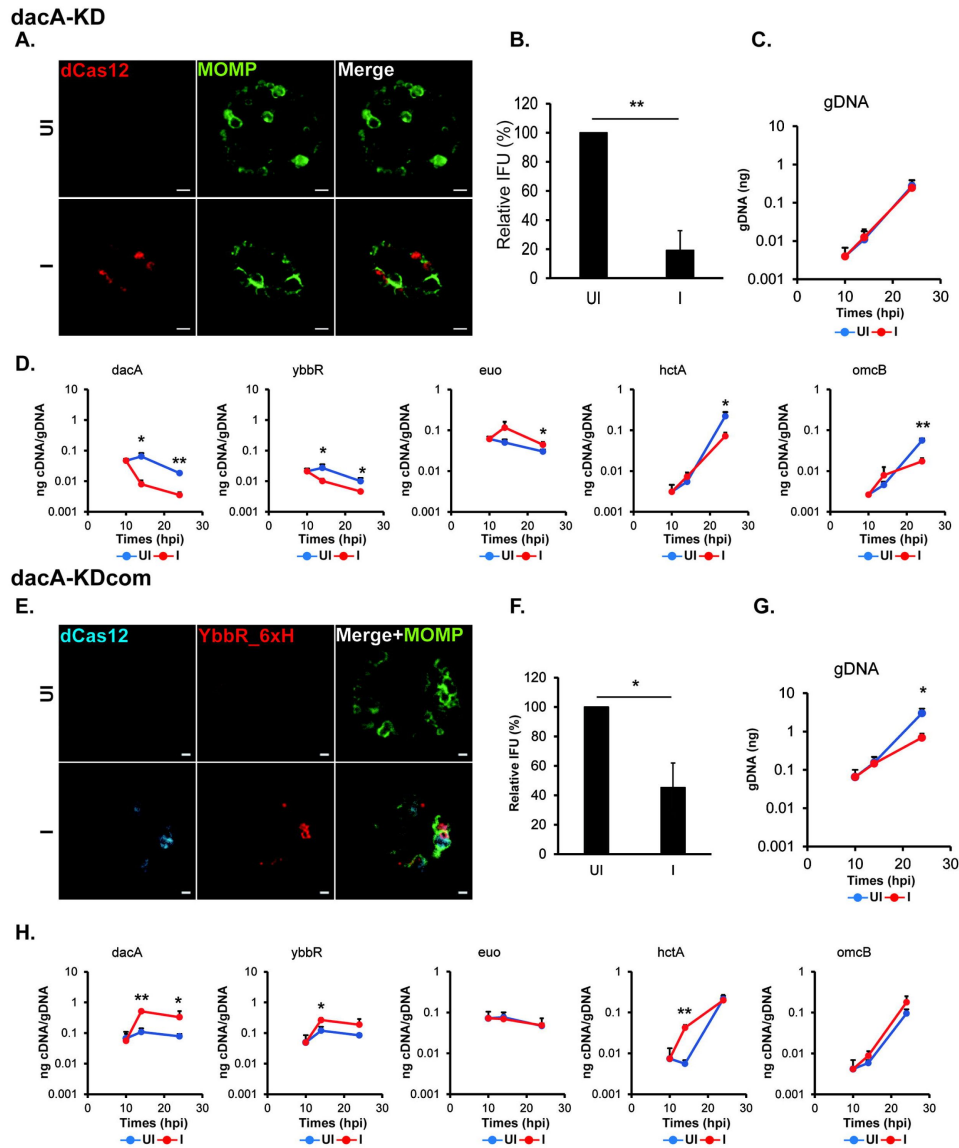


Figure 3.

CRISPRi-mediated *dacA-ybbR* knockdown displays reduced levels of transcripts for late genes.

(A-D) HeLa cells were infected with *C. trachomatis* transformed with a plasmid encoding an anhydrotetracycline (aTc)-inducible CRISPRi-dCas12 system targeting the *dacA* promoter (i.e., *dacA-KD*; see Fig. 1C). At 10 hpi, knockdown was induced or not with 5 nM aTc, and DNA and RNA samples were collected at 10, 14, and 24 hpi. Immunofluorescence analysis (IFA) and inclusion forming units (IFU) samples were collected at 24 hpi. **A.** IFA images of the *dacA-KD* strain at 24 hpi. Shown are individual panels of a representative inclusion for dCas12 and major outer membrane protein (MOMP) labeling as well as the merged image. **B.** Quantification of IFUs from uninduced and induced samples at 24 hpi. **C.** Quantification of genomic DNA by qPCR in uninduced and induced samples. **D.** Quantification of transcripts by RT-qPCR for *dacA*, *ybbR*, *euo*, *hctA*, and *omcB* from uninduced and induced samples. (E-H) HeLa cells were infected with *C. trachomatis* transformed with a plasmid encoding an aTc-inducible CRISPRi-dCas12 system targeting the *dacA* promoter and the complementing *dacA* and *ybbR_6xH* alleles (i.e., *dacA-KDcom*; see Fig. 1C). At 10 hpi, simultaneous knockdown and complementation were induced or not with 5 nM aTc, and immunofluorescence analysis (IFA), inclusion forming units (IFU), DNA, and RNA samples were collected as above. **E.** IFA images of *dacA-KD-dacAop* strain at 24 hpi. **F.** Quantification of IFU from uninduced and induced samples at 24 hpi. **G.** Quantification of genomic DNA by qPCR in uninduced and induced samples. **H.** Quantification of transcripts by RT-qPCR for *dacA*, *ybbR*, *euo*, *hctA*, and *omcB*. IFA images were acquired on a Zeiss AxioImager.Z2 equipped with an Apotome2 using a 100X lens objective. Scale bar: 1 μ m. UI = uninduced (i.e., -aTc); I = induced (i.e., +aTc) for all sample types. *: $p < 0.05$; **: $p < 0.001$ via two sample equal variance t-test.

Plugin of ImageJ; values near 1 indicate colocalization (32, 33)). This is not surprising as both proteins are critical for c-di-AMP synthesis based on our c-di-AMP measurements (Fig. 1D). Organism and inclusion morphology was similar to the *dacA*-KDcom complemented strain. We next assessed whether the overexpressed *DacA* and *YbbR_6xH* affected replication by measuring gDNA levels. Again, the gDNA data closely phenocopied the complemented knockdown strain, showing a decrease at 24 hpi (Fig. 4B). We then quantified transcripts for the developmentally regulated genes *euo*, *hctA*, and *omcB* as well as for *dacA* and *ybbR* (Fig. 4C). Not surprisingly, *dacA* and *ybbR* transcripts were elevated at the timepoints assessed. Transcripts for *euo* were not statistically changed but trended higher at the 24 hpi timepoint whereas *omcB* transcripts were slightly, but not significantly, increased at 14 and 24 hpi. Once again, we observed that *hctA* transcripts were increased over 10-fold at 14hpi in the developmental cycle (Fig. 4C), similar to what we measured for the complemented knockdown strain (Fig. 3H). These data reinforce that elevated c-di-AMP levels (Fig. 1D) in these strains lead to increased expression of the “early” late gene *hctA* at an earlier timepoint (14 hpi) in the developmental cycle.

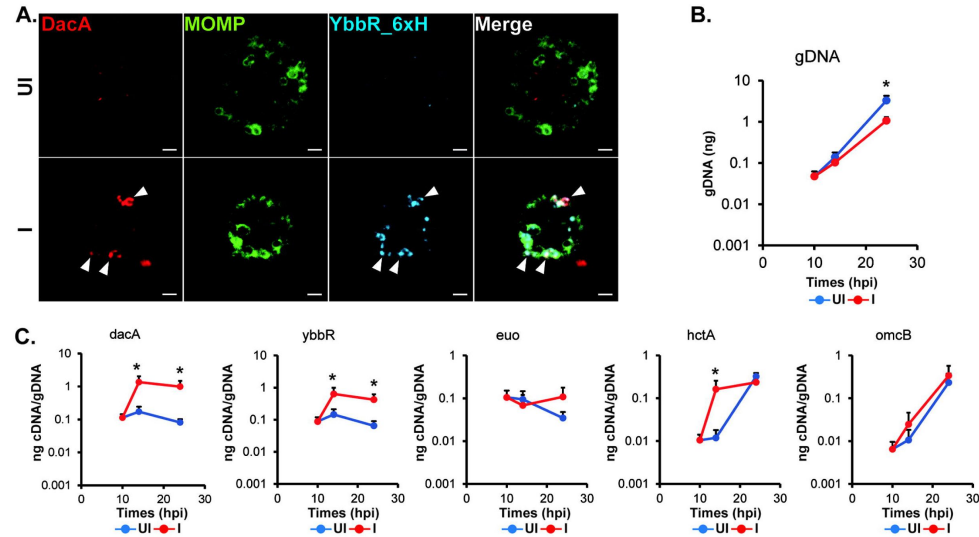
To confirm whether these phenotypes are caused by a higher level of c-di-AMP, we compared our results for the *dacAop* strain to the *dacAopMut* strain, in which the active site residue of *DacA* has been mutated. The c-di-AMP levels in this strain remain low (Fig. 1D). The bacterial morphology indicated slightly larger organisms without an effect on overall inclusion size (Fig. 4D). Genomic DNA levels showed a significant drop at 24 hpi (Fig. 4E). In assessing transcripts, *dacA* and *ybbR* were increased whereas *euo* levels were maintained, albeit not significantly so, at 24 hpi (Fig. 4F). In contrast to the c-di-AMP overproducing strains, transcripts for *hctA* and *omcB* were decreased at 24 hpi (Fig. 4F). The transcriptional results of the developmentally regulated genes in the *dacAopMut* strain were very similar to the *dacA*-KD strain (Fig. 3D). These data suggest that blocking c-di-AMP accumulation delays developmental cycle progression.

Elevated c-di-AMP increases transcript levels of genes necessary for secondary differentiation

Given the surprising finding that, in strains overproducing c-di-AMP, *hctA* transcripts were 10-fold higher at a timepoint not associated with secondary differentiation, we asked the question whether *all* genes related to secondary differentiation were increased after inducing production of c-di-AMP. Conversely, we wanted to explore whether reducing c-di-AMP levels would delay expression of genes related to secondary differentiation. Thus, to further investigate how c-di-AMP affects transcription of such genes, we performed RNA sequencing on both the *dacAop* overexpression and *dacA*-KD strains and compared the transcriptome between uninduced and induced samples within the given strain at the given timepoint. HeLa cells were infected with these transformants, and overexpression or knockdown was induced or not at 10 hpi with 5 nM aTc. For the *dacAop* strain, RNA was collected at 16 hpi, a time at which late genes are beginning to be expressed (as opposed to 14hpi) but remain near a basal level of transcription(12). The rationale for this was to determine whether high c-di-AMP levels result in increased late gene transcripts at this timepoint *above and beyond* the levels of the control, uninduced condition. For the *dacA*-KD strain, RNA was collected at 24 hpi, a time at which late genes are peaking in their expression. The rationale for this was to determine whether late gene transcription was decreased, which could not otherwise be assessed at the 16 hpi timepoint.

RNA sequencing results were statistically analyzed by the UNMC Bioinformatics Core (Table S2&3). Figure 5 shows a volcano plot of the results for the *dacAop* overexpression and *dacA*-KD strains. Of note, many late genes were evident in the upregulated quadrant for the *dacAop* strain whereas the opposite was true for the *dacA*-KD strain. We further characterized the upregulated or downregulated gene sets for the *dacAop* overexpression and *dacA*-KD strains, respectively, based on significant difference ($p < 0.05$) and fold-change (> 1.5). A summary of these results is presented in Table 1 (all data are presented in Table S3). We grouped the differentially expressed genes into five categories: 1) canonical late genes for which the literature has associated them with EB

dacAop



dacAopMut

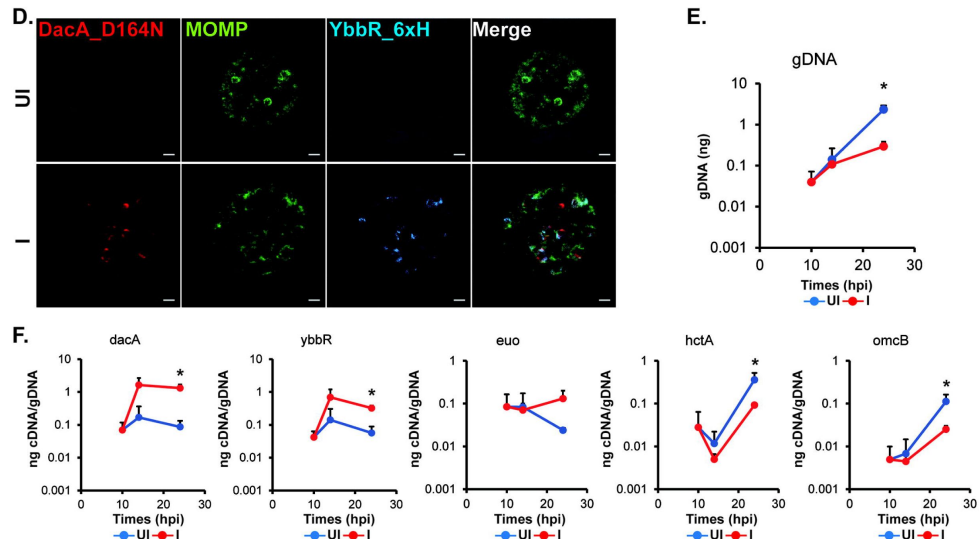


Figure 4.

Overexpression of DacA and YbbR_6xH prematurely increases *hctA* transcript levels.

(A-C) HeLa cells were infected with *C. trachomatis* transformed with an aTc-inducible plasmid encoding wild-type DacA and YbbR_6xH (i.e., *dacAop*; see Fig. 1C). At 10 hpi, expression of the constructs were induced or not with 5 nM aTc, and DNA and RNA samples were collected at 10, 14, and 24 hpi. Immunofluorescence analysis (IFA) samples were collected at 24 hpi. **A.** IFA images of the *dacAop* strain at 24 hpi. Shown are individual panels of a representative inclusion for DacA, YbbR_6xH, and major outer membrane protein (MOMP) labeling as well as the merged image. The arrowheads represent the co-localization of DacA and YbbR. **B.** Quantification of genomic DNA by qPCR in uninduced and induced samples. **C.** Quantification of transcripts by RT-qPCR for *dacA*, *ybbR*, *euo*, *hctA*, and *omcB*. **(D-F)** HeLa cells were infected with *C. trachomatis* transformed with an aTc-inducible plasmid encoding DacA(D164N) and YbbR_6xH (i.e., *dacAopMut*; see Fig. 1C). At 10 hpi, expression of the constructs were induced or not with 5 nM aTc, and DNA and RNA samples were collected at 10, 14, and 24 hpi. IFA samples were collected at 24 hpi. **D.** IFA images of *dacAopMut* strain at 24 hpi. Shown are individual panels for DacA, YbbR_6xH, and MOMP labeling as well as the merged image. **E.** Quantification of genomic DNA by qPCR in uninduced and induced samples. **F.** Quantification of transcripts by RT-qPCR for *dacA*, *ybbR*, *euo*, *hctA*, and *omcB* from uninduced and induced samples. IFA images were acquired on a Zeiss AxioImager.Z2 equipped with an Apotome2 using a 100X lens objective. Scale bar: 1 μ m. UI = uninduced (i.e., -aTc); I = induced (i.e., +aTc) for all sample types. *: $p < 0.05$; **: $p < 0.001$ via two sample equal variance t-test.

function, 2) outer membrane associated, 3) gene regulation associated, 4) glycogen synthesis associated, and 5) type III secretion system associated. Recent work from our group and the Hefty group to define the regulons of the alternative sigma factors in *Chlamydia* demonstrated that these sigma factors regulate some late gene expression associated with outer membrane remodeling, type III secretion, and other processes (34, 35). Consistent with our RT-qPCR data, we observed that all canonical late genes as well as all the other genes listed in these categories showed an increase in expression after c-di-AMP production was induced. In contrast, all the genes, and particularly the canonical late genes, showed a decrease in expression under conditions where c-di-AMP production was impaired. Overall, these RNA sequencing data confirm the direct influence of c-di-AMP on expression of genes related to secondary differentiation.

Increased c-di-AMP production activates late gene transcription during penicillin treatment

We next asked whether late gene transcripts could be increased by elevated c-di-AMP levels even under conditions when developmental progression is blocked. Penicillin treatment of *Chlamydia* blocks cell division and results in aberrantly enlarged RBs that fail to activate late gene transcription with a concomitant reduction in EB progeny production (36). As our previously described *dacAop* overexpression strain was constructed in a beta-lactamase producing background, we first generated a spectinomycin-resistant *dacAop* overexpression strain and validated that induction of the *dacA* operon resulted in earlier accumulation of *hctA* transcripts (Fig. S5). We next used this strain to infect cells and induced overexpression or not with 5 nM aTc at 10 hpi. We also treated cells with 1 U/mL penicillin at this timepoint. Penicillin treatment resulted in aberrantly enlarged RBs, as expected (Fig. 6A). However, we did observe that overexpressing *dacAop* in the presence of penicillin negatively impacted overall inclusion size as compared to the uninduced but penicillin-treated condition. Genomic DNA levels similarly were decreased at 24hpi as compared to the uninduced control (Fig. 6B). Transcripts for *dacA* and *ybbR* were increased after inducing their expression, and *euo* transcripts remained elevated at 24hpi under these conditions. Interestingly, *hctA* transcripts in the uninduced but penicillin-treated cultures were low at 14 hpi but increased ~5-fold by 24 hpi whereas *omcB* transcripts remained at a basal level at all timepoints under these conditions (Fig. 6C). Conversely, when *dacAop* is overexpressed in the presence of penicillin, *hctA* transcripts increased from 10 to 24hpi (Fig. 6C), albeit not to the same extent as measured in the absence of penicillin (Fig. 4). Transcripts for *omcB* also increased at 24hpi during *dacAop* overexpression in the presence of penicillin. These data suggest that, even under conditions when the developmental cycle is blocked, increased c-di-AMP levels can activate late gene transcription.

Alterations in c-di-AMP levels impact the timing of EB production

The phenotypic effect of increased late gene transcription should be an increase in EB production. However, if EB production is initiated at an earlier time in the developmental cycle when there are fewer non-infectious RBs to convert, then overall EB production should be decreased with a concomitant decrease in gDNA levels since only the RB replicates DNA. Conversely, delayed expression of late genes should be associated with a delay in EB production. Therefore, to test these predictions, we quantified EB production at 2h intervals from 18 to 24hpi to assess EB production during earlier phases of the developmental cycle, as well as at 32 and 48hpi to assess overall EB yields (Fig. 7). We did not detect any EBs at 16hpi or earlier (not shown). Cells were infected with the *dacAop*, *dacAopMut*, and *dacA-KD* strains and induced or not at 10 hpi with 5 nM aTc as previously noted. Consistent with the transcriptional data, we measured higher EB yields at 18 and 20 hpi during *dacAop* overexpression that quickly plateaued by 24 hpi (Fig. 7A). Conversely, when c-di-AMP accumulation was blocked by reducing the expression of DacA and YbbR (*dacA-KD*), we observed delayed EB production (Fig. 7B). We also measured lower EB yields when c-di-AMP levels were decreased by overexpression of the DacA(D164N) isoform and

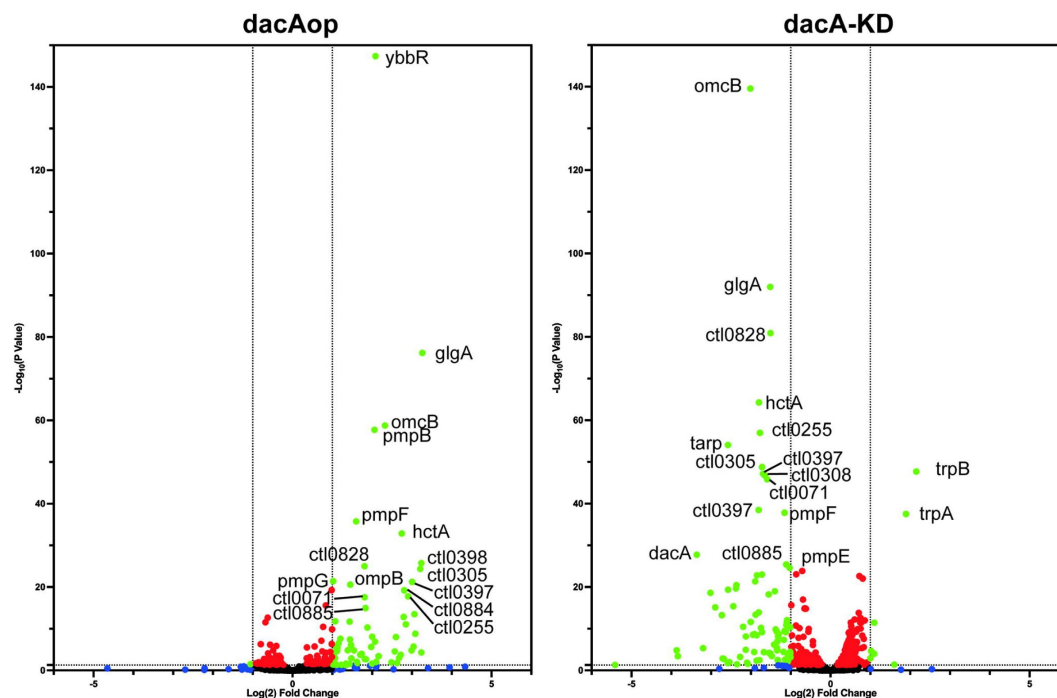


Figure 5.

The levels of c-di-AMP are correlated with late gene transcripts.

RNA sequencing was performed from HeLa cells infected with either the *dacAop* or *dacA-KD* strains. RNA samples were collected at 16hpi for *dacAop* and 24hpi for *dacA-KD* after inducing expression of the relevant constructs at 10hpi. Shown is a volcano plot of the RNA sequencing results with the vertical dashed lines indicating a two-fold change in transcript levels as compared to the respective uninduced control for the given strain and the horizontal lines indicating a p-value of 0.05. The plot was made using GraphPad Prism software. Green spots represent genes demonstrating a statistically significant two-fold change in transcription levels between uninduced and induced samples. Red dots represent genes with significant changes in transcript levels less than two-fold. Blue dots represent genes not significantly different but more than two-fold changed between the conditions. Black dots represent genes not significantly different and less than two-fold changed between the conditions. See also [Table 1](#) and Supplemental Tables 2 and 3 for more details.

			Canonical late genes	Fold change		
Gene ID	Ctrl D ORF	Name	Protein Names	<i>dacAop_OE</i>	<i>dacA-KD</i>	Ref.
CTL0112	CT743	<i>hctA</i>	Histone H1-like protein HC1	6.70	-3.47	(12, 56)
CTL0302	CT046	<i>hct2</i>	Histone H1-like protein HC2	3.52	-5.96	(12, 44)
CTL0336	CT080	<i>ltuB</i>	Late transcription unit B protein	4.00	-3.61	(12, 56)
CTL0700	CT441	<i>tsp</i>	Carboxy-terminal processing protease	6.18	-6.62	(44)
CTL0702	CT443	<i>omcB</i>	Large cysteine-rich periplasmic protein	5.00	-4.04	(12, 35)
CTL0703	CT444	<i>omcA</i>	Small cysteine-rich outer membrane protein	5.93	-3.71	(12, 35)
CTL0716	CT456	<i>tarp</i>	Translocated actin-recruiting phosphoprotein	6.06	-5.97	(34, 35)
			Membrane organization associated	Fold change		
Gene ID	Ctrl D ORF	Name	Protein Names	<i>dacAop_OE</i>	<i>dacA-KD</i>	Ref.
CTL0082	CT713	<i>ompB</i>	Outer membrane protein B	2.73	-2.15	(57)
CTL0248	CT869	<i>pmpE</i>	Polymorphic outer membrane protein	1.99	-2.04	(34, 57)
CTL0249	CT870	<i>pmpF</i>	Polymorphic outer membrane protein	3.03	-2.23	(34, 57)
CTL0250	CT871	<i>pmpG</i>	Polymorphic outer membrane protein	2.03	-1.55	(12, 57)
CTL0429	CT177	<i>dsbG</i>	Disulfide bond chaperone	2.79	-2.47	(34)
CTL0610	CT356	<i>dsbH</i>	Thioredoxin_DsbH domain-containing protein	3.07	-2.11	(12, 57)
CTL0670	CT413	<i>pmpB</i>	Polymorphic outer membrane protein	4.17	-1.74	(34, 57)
			Gene Regulation associated	Fold change		
Gene ID	Ctrl D ORF	Name	Protein Names	<i>dacAop_OE</i>	<i>dacA-KD</i>	Ref.
CTL0044	CT675	<i>mcsB</i>	Protein-arginine kinase	3.52	-2.21	(34)
CTL0045	CT676	<i>mcsA</i>	UVR domain-containing protein	2.53	-2.92	(34)
CTL0727	CT467	<i>atoS</i>	Two component regulator, histidine kinase	3.33	-1.80	(34)
CTL0728	CT468	<i>atoC</i>	Two component system response regulator	7.95	-2.78	
CTL0894	CT630	<i>chxR</i>	Atypical response regulator protein ChxR	3.68	-2.93	(58)
			Glycogen synthesis	Fold change		
Gene ID	Ctrl D ORF	Name	Protein Names	<i>dacAop_OE</i>	<i>dacA-KD</i>	Ref.
CTL0167	CT798	<i>glgA</i>	Glycogen synthase	9.58	-2.86	(12, 57)
CTL0342	CT087	<i>malQ</i>	4-alpha-glucanotransferase	9.43	-4.29	(41)
CTL0500	CT248	<i>glgP</i>	Alpha-1,4 glucan phosphorylase	2.70	-1.71	(57)
			Type III Secretion System	Fold change		
Gene ID	Ctrl D ORF	Name	Protein Names	<i>dacAop_OE</i>	<i>dacA-KD</i>	Ref.
CTL0041	CT672	<i>sctQ</i>	Type III secretion component, basal body	1.93	-1.76	(41)
CTL0043	CT674	<i>cdsC</i>	Type III secretion structural protein	1.79	-1.56	(41)
CTL0343	CT088	<i>sccI</i>	Type III secretion chaperone	4.48	-3.78	(41)
CTL0824	CT561	<i>sctL</i>	Type III secretion system protein	2.18	-1.75	(34)

Table 1.

Genes regulated by cyclic di-AMP levels.

***dacAop*(SpcR) + PenG**

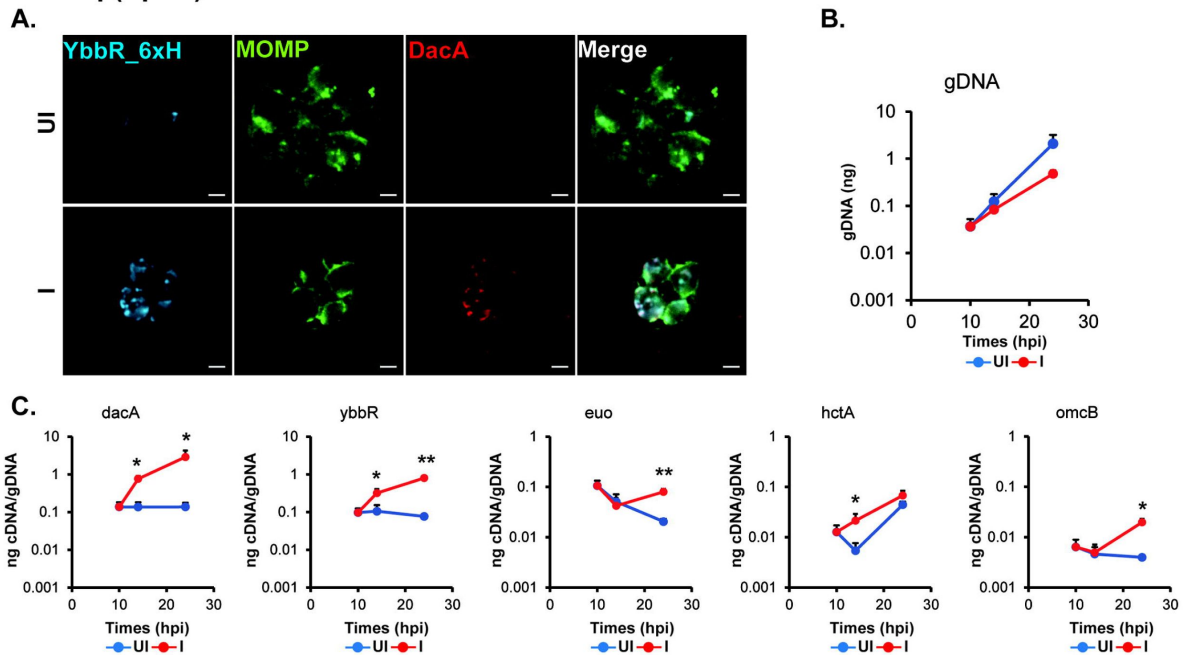


Figure 6.

Late genes transcripts are increased by high levels of c-di-AMP in penicillin-treated conditions.

HeLa cells were infected with a *C. trachomatis* strain transformed with a *dacAop* construct encoding spectinomycin resistance. At 10 hpi, expression of the construct was induced or not with 5 nM aTc and treated with 1 U/mL of penicillin G (PenG). DNA and RNA samples were collected at 10, 14, and 24 hpi, and immunofluorescence analysis (IFA) samples were collected at 24 hpi. **A.** IFA images of the *dacAop* strain in 1 U/mL PenG at 24 hpi. Shown are individual panels of a representative inclusion for DacA, YbbR_6xH, and major outer membrane protein (MOMP) labeling as well as the merged image. IFA images were acquired on a Zeiss AxioImager.Z2 equipped with an Apotome2 using a 100X lens objective. Scale bar: 2 μ m. **B.** Quantification of genomic DNA by qPCR in uninduced and induced samples. **C.** Quantification of transcripts by RT-qPCR for *dacA*, *ybbR*, *euo*, *hctA*, and *omcB*. UI = uninduced (i.e., -aTc); I = induced (i.e., +aTc) for all sample types. *: $p < 0.05$; **: $p < 0.001$ via two sample equal variance t-test.

YbbR (*dacAopMut*; **Fig. 7C**). The uninduced control conditions for each strain showed a steady accumulation of EBs throughout the course of the experiment, as expected. The vector control strain expressing mCherry showed no differences (Fig. S2). From these data, we conclude that earlier production of c-di-AMP levels results in earlier production of EBs.

Discussion

Secondary differentiation is essential for the propagation and survival of *Chlamydia* species. However, the mechanisms governing this essential process are largely unknown. It is possible that both internal and external environmental changes might serve as signals to trigger the shift from the non-infectious RB to the infectious EB. It is also possible, and even likely, that *Chlamydia* integrates multiple signal inputs during the differentiation process. However, as secondary differentiation is asynchronous, any mechanism must account for the stochasticity within the population of RBs “considering” differentiating to EBs. Here, we provide the first evidence of a signal that can directly activate EB-associated gene expression with a concomitant early production of EBs.

Several reports have proposed mechanisms by which *Chlamydia* triggers secondary differentiation. For example, Thompson et al. investigated the non-canonical regulation of the major sigma factor, σ^{66} , by the Rsb phosphoregulatory system(37). The authors provided evidence that altering the levels of RsbW or RsbV1 impacted the expression of σ^{66} -controlled genes. This led them to propose a model whereby sensing of ATP by the Rsb system controls the availability of σ^{66} , implying that low levels of ATP lead to initiation of secondary differentiation. Further support from this came from work by Kuwabara et al. who showed effects of ATP, GTP, and glucose on the activity of different Rsb components(38). Finally, work from Soules et al. identified TCA intermediates as ligands for RsbU, linking the TCA cycle and ATP synthesis to the Rsb system(39). However, no effect was noted on premature late gene expression in these experimental systems.

Sequestration/inactivation of the major sigma factor under low ATP conditions would presumably render RNA polymerase free to interact with the alternative sigma factors, σ^{54} and σ^{28} . These alternative sigma factors have been linked to late gene expression with studies from Soules et al. and Hatch and Ouellette indicating that σ^{54} is associated with increased transcription of outer membrane components, type III secretion (T3S) system components, and other genes typically expressed late in development(34, 35). However, some of these changes are likely indirect and downstream to the σ^{54} regulon itself since some affected genes have been shown to have σ^{66} promoter sequence elements(40, 41). Significant remodeling of the outer membrane occurs as the RB transitions to the EB(42), and, similarly, the EB is prepackaged with T3S effectors, like TarP(43), that facilitate EB invasion into a target host cell. σ^{28} was shown to control expression of two canonical late genes: *hctB* and *tsp*(34, 44). Each of the encoded proteins is highly toxic if overexpressed(45, 46), suggesting that the additional layer of regulation by σ^{28} is necessary to ensure EBs are formed at the correct time. Hatch and Ouellette proposed that these genes may be the last to be activated precisely because of this(34). Overall, these data clearly link transcriptional regulation, via the sigma factors, to developmental cycle progression and secondary differentiation.

Despite the clear changes in transcription that occur during the chlamydial developmental cycle, activation of late gene expression alone does not guarantee secondary differentiation. We recently explored the function of the Clp protease systems in chlamydial growth and development(47). Interestingly, we observed that the inability to degrade SsrA-tagged products prevented functional secondary differentiation with a severe defect in EB production. This was demonstrated in strains unable to degrade SsrA products either by preventing their recognition by ClpX or by altering the

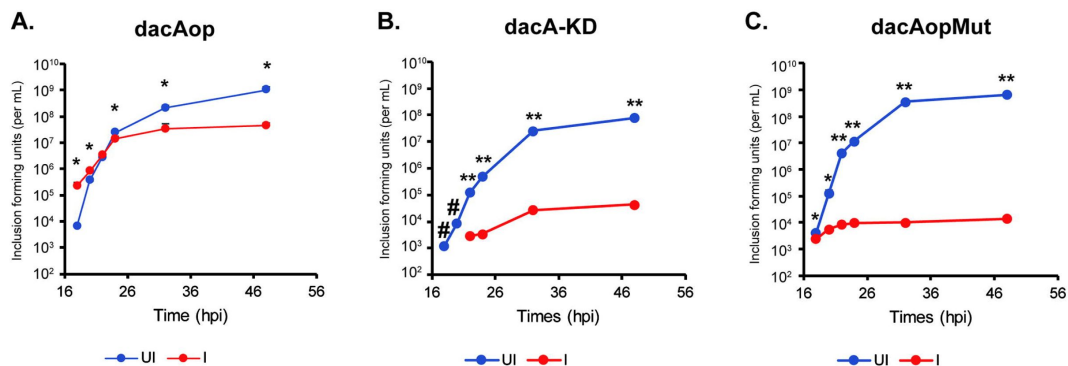


Figure 7.

EB production is induced by high levels of c-di-AMP.

(A-C) HeLa cells were infected with the chlamydial transformants *dacAop* (A), *dacA-KD* (B), and *dacAopMut* (C). Expression of the constructs was induced or not with 5 nM aTc at 10hpi. At 18, 20, 22, 24, 32, and 48 hpi, infected cell lysates were harvested for IFU quantification. UI = uninduced (i.e., -aTc); I = induced (i.e., +aTc) for all sample types. #: Non-detected from the induced samples. *: $p < 0.05$; **: $p < 0.001$ via two sample equal variance t-test.

SsrA tag to a version that is not degraded efficiently. Nonetheless, late gene transcription was activated in these strains. Therefore, post-translational regulation of secondary differentiation is also critical to the process.

Assuming a post-translational gene regulation model for secondary differentiation, we were interested in exploring known differences between EBs and RBs as a clue to what factors could be serving as a signal for this process. EBs and RBs differ in a number of characteristics, from their function to their morphology to their gene expression. In 2013, a study from Barker et al. added another difference – their relative levels of c-di-AMP with RBs having lower levels than EBs. Although the study from Barker et al. was focused on understanding how IFN β is activated in *Chlamydia*-infected cells, we were intrigued by this relative difference in c-di-AMP levels. We hypothesized that c-di-AMP production might be a signal for secondary differentiation. We compared chlamydial growth in normal HeLa or STING-KO HeLa cells but could not detect any differences in growth in these cell types (Fig. 2 and S2 vs Fig. S6), indicating that c-di-AMP production by *Chlamydia* is unlikely used as means of activating the host cell. Rather, STING activation is “accidental”. Using recently developed genetic tools for *Chlamydia*, we demonstrated that we could manipulate the levels of c-di-AMP in the bacterium to either block or stimulate its production. Excitingly, we observed that elevated c-di-AMP was linked to earlier secondary differentiation whereas blocking c-di-AMP production prevented EB production (Fig. 6). Importantly, c-di-AMP production resulted in an increase in late gene transcripts as noted above for the function of the alternative sigma factors. Therefore, c-di-AMP may act as a post-translational mechanism to trigger secondary differentiation. However, we cannot exclude a function for c-di-AMP in directly manipulating gene expression by riboswitches (48), and further work is necessary to understand how c-di-AMP directly controls chlamydial development. To our knowledge, this is the first study to identify and provide experimental evidence for a signaling factor involved in differentiation of an obligate intracellular bacterium.

How might c-di-AMP function in *Chlamydia*? This is not readily apparent because *Chlamydia* lacks orthologs of proteins that have been characterized in other bacterial systems to bind c-di-AMP. For example, *Chlamydia* lacks annotated K⁺ transporters, and it is unknown how these bacteria maintain their osmolarity. However, a recent study suggested that changes in K⁺ levels affect the chlamydial developmental cycle (49). Therefore, c-di-AMP may function as a K⁺ homeostasis coordinator in *Chlamydia* through as-yet unknown pathways. We did observe altered chlamydial morphology in some of our strains in which we altered c-di-AMP levels, and this may indicate changes in osmostability or effects on cell division. In regard to the latter possibility, in *S. aureus*, DacA activity is inhibited by interacting with GlmM (phosphoglucosamine mutase) (50). GlmM produces UDP-GlcNAc that is a precursor in both peptidoglycan and LPS synthesis. Given that *Chlamydia* uses peptidoglycan for cell division, we tested whether DacA and YbbR were localized to the division septum. However, their localization was not associated with the division septum (Fig. S7). In addition, we did not detect an interaction between DacA and GlmM using the Bacterial Adenylate Cyclase-based Two Hybrid (BACTH) system (data not shown), and the chlamydial GlmM lacks the residues required for DacA interaction. Therefore, we conclude that DacA and YbbR are unlikely to directly impact cell division. However, based on the cell morphological changes in DacA overexpression, we cannot exclude a function for c-di-AMP on cell wall metabolism. Future studies will focus on identifying c-di-AMP binding proteins and characterizing their function in *Chlamydia*.

The use of c-di-AMP as a signal for secondary differentiation presents a “chicken-or-egg” quandary. How is the activity and expression of DacA, the diadenylate cyclase, regulated? Our transcript analyses indicate that *dacA-ybbR* transcripts peak during the RB phase of growth, similar to most genes in *Chlamydia*. One possibility not easily tested due to the obligate intracellular nature of *Chlamydia*, is that c-di-AMP production depletes ATP levels, which then reduce σ^{66} activity as described above. Our data indicate YbbR is necessary to activate DacA function since expressing DacA alone had no significant impact on overall c-di-AMP levels (Fig. 1D). However,

overexpressing DacA alone did negatively impact chlamydial growth, suggesting that the balance between DacA and YbbR levels is important, with YbbR being the limiting factor for c-di-AMP production. For example, too much DacA insertion into the membrane without binding to YbbR may disrupt the membrane biology of the RB. Therefore, YbbR itself may be a target for regulation. As a monotopic transmembrane protein, YbbR may be degraded by proteases such as the inner membrane-associated FtsH or the periplasmic proteases HtrA or Tsp. This would presumably shut down c-di-AMP synthesis. Alternatively, DacA may interact with other membrane proteins, and overexpressing it alone may impair the function of this binding partner(s), resulting in the observed phenotypes. However, this possibility contradicts the effects of overexpressing the D164N isoform, which blocked c-di-AMP production. These experiments were conducted in the presence of the chromosomally expressed copy of DacA, and we suggest that the mutant isoform acts as a dominant negative by interfering with the wild-type function or by titrating away YbbR from the wild-type DacA. In contrast, knocking down *dacA-ybbR* transcripts will reduce, but not completely eliminate, DacA-YbbR protein, which may result in residual c-di-AMP levels as compared to the DacA mutant as well as the slight phenotypic differences between these strains. Further work is needed to test these possibilities. However, our data are clear in linking c-di-AMP levels to chlamydial developmental progression.

Even though diverse functions of c-di-AMP in other bacteria have been reported previously, this is the first time c-di-AMP has been described as a checkpoint in chlamydial development. It is possible that the levels of c-di-AMP act as a de facto means for monitoring the bacterial population within the inclusion or for their overall developmental status. Canonical bacteria can use quorum sensing to detect bacteria at the population level and to communicate with other species⁽⁵¹⁾. However, *C. trachomatis* lacks homologues of genes related to quorum sensing⁽²⁸⁾. In addition, secondary differentiation occurs asynchronously, and this feature is different from a quorum sensing mechanism. Our model accounts for the asynchronicity of secondary differentiation. As RBs accumulate c-di-AMP and divide, it is likely the c-di-AMP will be distributed unevenly between the mother and daughter cell since *Chlamydia* divides through an asymmetric budding mechanism⁽⁷⁾ (**Fig. 8**). This will lead to two cells with different levels of c-di-AMP, one of which may then accumulate enough c-di-AMP to trigger secondary differentiation while the other continues to divide (**Fig. 8**). This is consistent with a recent model proposing that a population of division-competent RBs is maintained throughout the developmental cycle⁽⁵²⁾.

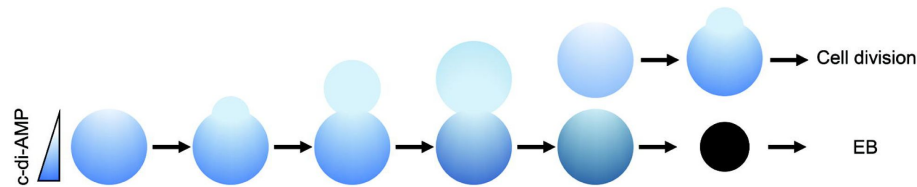
Other ongoing studies in the lab have revealed additional post-translational mechanisms that drive secondary differentiation (Jensen et al. *in revision*; ⁽⁵³⁾), thus it seems likely that *Chlamydia* integrates multiple signal inputs to ensure that all conditions are met before undergoing this essential step in their biology.

Materials and Methods

Organisms and Cell Culture

Wild-type (ATCC, Manassas, VA) and STING KO (Dr. Frank van Kuppeveld; Utrecht University) HeLa, human cervical epithelial-derived, and McCoy (kind gift of Dr. Harlan Caldwell, NIH), mouse fibroblast-derived, cell lines were cultured at 37°C with 5% CO₂ in Dulbecco's Modified Eagle Medium (DMEM; Invitrogen, Waltham, MA) containing 10% fetal bovine serum (FBS; Hyclone, Logan, UT) and 10 µg/mL gentamicin (Gibco, Waltham, MA). *C. trachomatis* serovar L2 (434/Bu) lacking the endogenous plasmid (-pL2; kind gift of Dr. Ian Clarke, Univ. Southampton) was used for transformation. All cell cultures and chlamydial stocks were routinely tested for Mycoplasma contamination using the Mycoplasma PCR detection kit (Sigma, St. Louis, MO). For *E. coli*, NEB10β competent cells (New England Biolabs, Ipswich, MA) were used for the amplification of pBOMB-derivative vectors. *E. coli* was grown at 30°C in LB media. All chemicals and antibiotics were obtained from Sigma unless otherwise noted.

A.



B.

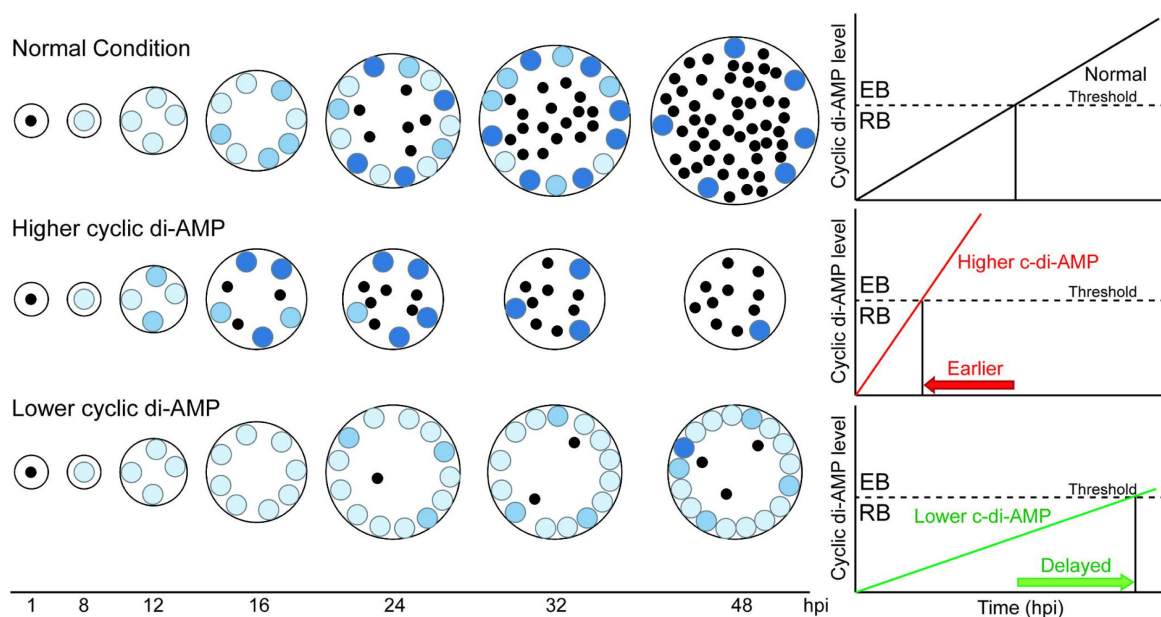


Figure 8.

A working model illustrating how c-di-AMP impacts the onset of secondary differentiation in *C. trachomatis*.

A. During RB division, there is a gradient of c-di-AMP that forms between the mother and daughter cell. The lighter blue represents lower levels of c-di-AMP, the darker blue represents higher levels of c-di-AMP. In the depicted scenario, the daughter cell is able to divide again, having not reached a critical threshold of c-di-AMP. In contrast, the mother cell accumulates sufficient c-di-AMP to trigger secondary differentiation to an EB. **B.** A model illustrating how c-di-AMP levels impact secondary differentiation progression. Shown is the “normal” condition of wild-type bacteria, bacteria overexpressing DacA and YbbR (higher cyclic di-AMP), and bacteria expressing the CRISPRi system targeting the *dacA* promoter (lower cyclic di-AMP). Depending on the amount of c-di-AMP produced, secondary differentiation can either be triggered earlier or later as shown. The black dots represent EBs; the bigger and blue-colored circles show the RB cells. The brightness of blue reflects the c-di-AMP level.

Cloning

The list of the vectors and primers used in this study is detailed in Supplemental Table 1. Target genes were amplified by PCR with Phusion DNA polymerase (NEB) using 10 ng *C. trachomatis* L2 genomic DNA or appropriate vectors as a template. Some DNA segments were directly synthesized as a gBlock fragment (Integrated DNA Technologies, Coralville, IA). If plasmids were used as a template, then we treated the PCR product with DpnI enzyme to remove templates. The PCR products were purified using a PCR purification kit (Qiagen, Hilden, Germany). The HiFi Assembly reaction master mix (NEB) was used following the manufacturer's manual in conjunction with plasmids pBOMBL(30) linearized with EagI and KpnI or pBOMBL12CRia (empty vector) linearized with BamHI. The linearized plasmids were also dephosphorylated with FastAP (ThermoFisher). The products of the HiFi reaction were transformed into NEB10 β competent cells (NEB) and plated on LB agar with appropriate antibiotics. Plasmids were subsequently isolated using a mini-prep kit (Qiagen) and verified by plasmid digest and sequencing from individual colonies grown overnight in LB broth with appropriate antibiotic selection.

Transformation of *Chlamydia trachomatis*

McCoy cells were plated in a six-well plate the day before beginning the transformation procedure. *C. trachomatis* serovar L2 without plasmid (-pL2) resuspended with Tris-CaCl₂ buffer (10 mM Tris-Cl pH 7.5, 50 mM CaCl₂) was incubated with 2 μ g plasmid at room temperature for 30 min. During this step, McCoy cells were washed with 2 mL Hank's Balanced Salt Solution (HBSS) media containing Ca²⁺ and Mg²⁺ (Gibco). After that, McCoy cells were infected with the transformants in 2 mL HBSS per well. The plate was centrifuged at 400 x g for 15 min at room temperature and incubated at 37°C for 15 min. The inoculum was aspirated, and 2 mL 1X DMEM containing 10% FBS and 10 μ g/mL gentamicin was added per well. At 8 hours post infection (hpi), 1 μ g/mL cycloheximide and either 1 or 2 U/mL penicillin G or 500 μ g/mL spectinomycin were added, and the plate was incubated at 37°C until 48 hpi. At 48 hpi, the transformants were harvested and infected onto a new McCoy cell monolayer. These harvest and infection steps were repeated every 44-48 hpi until fluorescent, antibiotic-resistant inclusions were observed.

(RT-)qPCR

HeLa cells were infected with chlamydial transformants at an MOI of 0.5. At 10 hpi, 5 nM anhydrotetracycline (aTc) was added or not to the culture medium. Total RNA and DNA were harvested at this time point from duplicate wells not treated with aTc. At 14 and 24 hpi, total RNA and DNA were collected using Trizol (Invitrogen) and DNeasy Tissue (Qiagen) kit, respectively, as described elsewhere(34). After DNase treatment of total RNA, cDNA was synthesized using Superscript III reverse transcriptase (Invitrogen). After diluting the cDNA 10-fold, 5 μ L of the diluted cDNA was used as a template for qPCR. Equal masses of genomic DNA were used from each of the samples to quantify chlamydial genomes, which were used to normalize transcript data as described(40). For both cDNA and gDNA samples, qPCR reactions were prepared using 2X SYBR Green (ThermoFisher) in a total volume of 25 μ L per well. Standard cycling conditions were used with a melting curve analysis to verify products. Transcripts and genome copies were assessed from at least three biological replicates.

Indirect immunofluorescence assay (IFA)

HeLa cells were infected with chlamydial transformants as above. At 10 hpi, 5 nM aTc was added or not, and the infected cells were fixed with fixing solution (3.2% formaldehyde and 0.022% glutaraldehyde in 1X DPBS) for 2 min and permeabilized with 90% MeOH for 1 min at 10.5 hpi or 24 hpi. The fixed cells were labeled with primary antibodies including rabbit anti-DacA (custom anti-peptide antibody targeting the C-terminal TRNERKTNPISWMRKK prepared by Pacific Immunology, Ramona, CA), goat anti-major outer-membrane protein (MOMP; Meridian, Memphis, TN), and rabbit anti-six histidine tag (Genscript, Piscataway, NJ and Abcam, Cambridge, UK,

respectively). To visualize the primary antibodies, donkey anti-goat antibody (488), donkey anti-mouse (405), or donkey anti-rabbit antibody (594) were used as secondary antibodies. The secondary antibodies were obtained from Invitrogen or Jackson Immunology (West Grove, PA). Coverslips were observed using a Zeiss AxioImager.Z2 with Apotome2 as noted in the figure legends.

Inclusion forming unit (IFU) measurement

HeLa cells were infected with chlamydial transformants as above. At 10 hpi, 5 nM aTc was added or not to the culture medium. At the indicated times, infected cells were harvested in 1 mL 2SP media then frozen at -80°C . After thawing the lysates, the samples were serially 1:10 diluted and used to infect HeLa cells seeded in 24-well plates. At 24 hpi, the number of GFP expressing inclusions was counted from 30 fields of view to calculate the IFUs from the original sample. Three biological replicates were performed.

Cyclic di-AMP measurement

HeLa cells were infected with the indicated chlamydial transformants as above. At 10 hpi, expression of the constructs was induced or not with 5 nM aTc, and, at 16 and 24 hpi, samples were prepared for measuring the level of cyclic di-AMP. After aspirating the media, the infected cell monolayers were washed with 1X PBS and resuspended with B-Per Bacterial Cell Lysis Buffer (Thermo). Samples were vortexed for 1 min and then centrifuged at 13,300 $\times$ g for 15 min at 4°C . The levels of cyclic di-AMP from the supernatants of the cell lysates were quantified using the Cyclic di-AMP ELISA kit (Cayman) following the manufacturer's instructions(54 [4](#)).

Preparation of RNA sequencing samples

Samples were prepared as reported previously(34 [4](#)). Briefly, the transformants of *dacAop* and *dacA-KD* were infected into HeLa cells. At 10 hpi, the constructs were induced or not with 5 nM aTc. RNA samples were prepared from *dacAop*-infected cells and *dacA-KD*-infected cells at 16 hpi and 24 hpi, respectively. 20 μg RNA samples were treated with DNase to remove DNA contamination using DNA-free Turbo kit (Thermo) according to the manufacturer's instruction. Ribosomal RNA was depleted from samples using the MICROBEnrich (Thermo) and MICROBExpress kits (Thermo) following the manufacturer's instructions. RNA samples were processed for sequencing by the UNMC Genomics Core Facility. The resultant libraries from the individual samples were multiplexed and subjected to 100-bp paired-read sequencing to generate approximately 60 million pairs of reads per sample on an Illumina NovaSeq6000 sequencer in the UNMC Genomics Core facility. The original fastq format reads were trimmed by fqtrim tool (<https://ccb.jhu.edu/software/fqtrim> [4](#)) to remove adapters, terminal unknown bases (Ns) and low quality 3' regions (Phred score < 30). The trimmed fastq files were processed by FastQC (55 [4](#)) for quality control. *Chlamydia trachomatis* 434/Bu bacterial reference genome and annotation files were downloaded from Ensembl (http://bacteria.ensembl.org/Chlamydia_trachomatis_434_bu/Info/Index [4](#)). Sequencing data were analyzed by the Bioinformatics and Systems Biology Core (BSBC). The trimmed fastq files were mapped to *Chlamydia trachomatis* 434/Bu by CLC Genomics Workbench 23 for RNAseq analyses.

Data availability

The raw and processed RNA sequencing reads in fastq format have been deposited in the Gene Expression Omnibus (GEO; www.ncbi.nlm.nih.gov/geo/ [4](#)) under accession no. GSE252732.

Statistical Analysis

To analyze the statistical significance between uninduced and induced samples of qPCR and IFU data, we used two sample equal variance Student's t-test. For the levels of cyclic di-AMP data, we used a one-way ANOVA followed by Tukey's posthoc comparison or a ratio paired t-test.

Data and materials availability

All data are available in the main text or the supplementary materials.

Acknowledgements

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Additional information

Author contributions

Junghoon Lee, Methodology, Investigation, Visualization, Writing – original draft, and Writing-review & editing

Scot P. Ouellette, Conceptualization, Methodology, Investigation, Visualization, Supervision, and Writing-review & editing

Additional files

Figure S1 [↗](#)

Figure S2 [↗](#)

Figure S3 [↗](#)

Figure S4 [↗](#)

Figure S5 [↗](#)

Figure S6 [↗](#)

Figure S7 [↗](#)

Supplemental Figure Legends [↗](#)

Supplemental Table S1 [↗](#)

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Author information

Junghoon Lee

Department of Pathology, Microbiology, and Immunology, College of Medicine, University of Nebraska Medical Center, Omaha, United States

Scot P Ouellette

Department of Pathology, Microbiology, and Immunology, College of Medicine, University of Nebraska Medical Center, Omaha, United States

ORCID iD: [0000-0002-3721-6839](https://orcid.org/0000-0002-3721-6839)

For correspondence: scot.ouellette@unmc.edu

Editors

Reviewing Editor

Bavesh Kana

University of the Witwatersrand, Johannesburg, South Africa

Senior Editor

Bavesh Kana

University of the Witwatersrand, Johannesburg, South Africa

Reviewer #1 (Public review):

Summary:

The paper by Lee and Ouellette explores the role of cyclic-d-AMP in chlamydial developmental progression. The manuscript uses a collection of different recombinant plasmids to up- and down-regulate cdAMP production, and then uses classical molecular and microbiological approaches to examine the effects of expression induction in each of the transformed strains.

Strengths:

This laboratory is a leader in the use of molecular genetic manipulation in *Chlamydia trachomatis* and their efforts to make such efforts mainstream is commendable. Overall, the model described and defended by these investigators is thorough and significant.

Weaknesses:

The biggest weakness in the document is their reliance on quantitative data that is statistically not significant, in the interpretation of results. These challenges can be addressed in a revision by the authors.

<https://doi.org/10.7554/eLife.104240.1.sa2>

Reviewer #2 (Public review):

Summary:

This manuscript describes the role of the production of c-di-AMP on the chlamydial developmental cycle. Chlamydia are obligate intracellular bacterial pathogens that rely on eukaryotic host cells for growth. The chlamydial life cycle depends on a cell form developmental cycle that produces phenotypically distinct cell forms with specific roles during the infectious cycle. The RB cell form replicates amplifying chlamydia numbers while the EB cell form mediates entry into new host cells disseminating the infection to new hosts. Regulation of cell form development is a critical question in chlamydia biology and pathogenesis. Chlamydia must balance amplification (RB numbers) and dissemination (EB numbers) to maximize survival in its infection niche. The main findings In this manuscript show that overexpression of the *dacA-ybbR* operon results in increased production of c-di-AMP and early expression of the transitionary gene *hctA* and late gene *omcB*. The authors also knocked down the expression of the *dacA-ybbR* operon and reported a reduction in the expression of both *hctA* and *omcB*. The authors conclude with a model suggesting the amount of c-di-AMP determines the fate of the RB, continued replication, or EB conversion. Overall, this is a very intriguing study with important implications however the data is very preliminary and the model is very rudimentary and is not well supported by the data.

Describing the significance of the findings:

The findings are important and point to very exciting new avenues to explore the important questions in chlamydial cell form development. The authors present a model that is not quantified and does not match the data well.

Describing the strength of evidence:

The evidence presented is incomplete. The authors do a nice job of showing that overexpression of the *dacA-ybbR* operon increases c-di-AMP and that knockdown or overexpression of the catalytically dead *DacA* protein decreases the c-di-AMP levels. However, the effects on the developmental cycle and how they fit the proposed model are less well supported.

dacA-ybbR ectopic expression:

For the *dacA-ybbR* ectopic expression experiments they show that *hctA* is induced early but there is no significant change in *OmcB* gene expression. This is problematic as when RBs are treated with Pen (this paper) and (DOI 10.1128/MSYSTEMS.00689-20) *hctA* is expressed in the aberrant cell forms but these forms do not go on to express the late genes suggesting stress events can result in changes in the developmental expression kinetic profile. The RNA-seq data are a little reassuring as many of the EB/Late genes were shown to be upregulated by *dacA-ybbR* ectopic expression in this assay.

The authors also demonstrate that this ectopic expression reduces the overall growth rate but produces EBs earlier in the cycle but overall fewer EBs late in the cycle. This observation matches their model well as when RBs convert early there is less amplification of cell numbers.

dacA knockdown and *dacA*(mut)

The authors showed that *dacA* knockdown and ectopic expression of the *dacA* mutant both reduced the amount of c-di-AMP. The authors show that for both of these conditions, *hctA* and *omcB* expression is reduced at 24 hpi. This was also partially supported by the RNA-seq data for the *dacA* knockdown as many of the late genes were downregulated. However, a shift to an increase in RB-only genes was not readily evident. This is maybe not surprising as the chlamydial inclusion would just have an increase in RB forms and changes in cell form ratios would need more time points.

Interestingly, the overall growth rate appears to differ in these two conditions, growth is unaffected by *dacA* knockdown but is significantly affected by the expression of the mutant. In both cases, EB production is repressed. The overall model they present does not support this data well as if RBs were blocked from converting into EBs then the growth rate should increase as the RB cell form replicates while the EB cell form does not. This should shift the population to replicating cells.

Overall this is a very intriguing finding that will require more gene expression data, phenotypic characterization of cell forms, and better quantitative models to fully interpret these findings.

<https://doi.org/10.7554/eLife.104240.1.sa1>

Author response:

Public Reviews:

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The biggest weakness in the document is their reliance on quantitative data that is statistically not significant, in the interpretation of results. These challenges can be addressed in a revision by the authors.

Thank you for these comments. We have generated new data, which we hope the reviewer will find more compelling. These will be included in a revised manuscript.

Reviewer #2 (Public review):

Summary:

*This manuscript describes the role of the production of c-di-AMP on the chlamydial developmental cycle. Chlamydia are obligate intracellular bacterial pathogens that rely on eukaryotic host cells for growth. The chlamydial life cycle depends on a cell form developmental cycle that produces phenotypically distinct cell forms with specific roles during the infectious cycle. The RB cell form replicates amplifying chlamydia numbers while the EB cell form mediates entry into new host cells disseminating the infection to new hosts. Regulation of cell form development is a critical question in chlamydia biology and pathogenesis. Chlamydia must balance amplification (RB numbers) and dissemination (EB numbers) to maximize survival in its infection niche. The main findings in this manuscript show that overexpression of the *dacA-ybbR* operon results in*

increased production of c-di-AMP and early expression of the transitional gene hctA and late gene omcB. The authors also knocked down the expression of the dacA-ybbR operon and reported a reduction in the expression of both hctA and omcB. The authors conclude with a model suggesting the amount of c-di-AMP determines the fate of the RB, continued replication, or EB conversion. Overall, this is a very intriguing study with important implications however the data is very preliminary and the model is very rudimentary and is not well supported by the data.

Thank you for your comments. *Chlamydia* is not an easy experimental system, but we will do our best to address the reviewer's concerns in a revised submission.

Describing the significance of the findings:

The findings are important and point to very exciting new avenues to explore the important questions in chlamydial cell form development. The authors present a model that is not quantified and does not match the data well.

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The evidence presented is incomplete. The authors do a nice job of showing that overexpression of the dacA-ybbR operon increases c-di-AMP and that knockdown or overexpression of the catalytically dead DacA protein decreases the c-di-AMP levels. However, the effects on the developmental cycle and how they fit the proposed model are less well supported.

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As the reviewer notes, we also generated RNAseq data, which validates that late gene transcripts (including sigma28 and sigma54 regulated genes) are statistically significantly increased earlier in the developmental cycle in parallel to increased c-di-AMP levels. The lack of statistical significance in the RT-qPCR data for omcB, which shows a trend of higher transcripts, is less concerning given the statistically significantly RNAseq dataset. We have reported the data from three replicates for the RT-qPCR and do not think it would be worthwhile to attempt more replicates in an attempt to "achieve" statistical significance.

The authors also demonstrate that this ectopic expression reduces the overall growth rate but produces EBs earlier in the cycle but overall fewer EBs late in the cycle. This observation matches their model well as when RBs convert early there is less amplification of cell numbers.

dacA knockdown and dacA(mut)

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This is maybe not surprising as the chlamydial inclusion would just have an increase in RB forms and changes in cell form ratios would need more time points.

Thank you for this comment. We agree that it is not surprising given the shift in cell forms. The reduction in *hctA* transcripts argues against a stress state as noted above by the reviewer, and the RNAseq data from *dacA*-KD conditions indicates at least that secondary differentiation has been delayed. We will try to clarify this in a revision.

*Interestingly, the overall growth rate appears to differ in these two conditions, growth is unaffected by *dacA* knockdown but is significantly affected by the expression of the mutant. In both cases, EB production is repressed. The overall model they present does not support this data well as if RBs were blocked from converting into EBs then the growth rate should increase as the RB cell form replicates while the EB cell form does not. This should shift the population to replicating cells.*

We agree that it seems that perturbing c-di-AMP production, whether by knockdown or overexpressing the mutant *DacA*(D164N), has an overall negative impact on chlamydial growth. We have generated new data, which we think will address this. These new data will be included in a revised manuscript.

Overall this is a very intriguing finding that will require more gene expression data, phenotypic characterization of cell forms, and better quantitative models to fully interpret these findings.

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